

QUATERNARY HISTORY OF DECIDUOUS FORESTS OF EASTERN NORTH AMERICA AND EUROPE¹

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ABSTRACT

The temperate deciduous forest of North America is more diverse than the deciduous forest of western Europe. This difference has traditionally been explained by greater survival in North America of deciduous species during the Quaternary. More recent investigations have shown, however, that late-Tertiary forests of Europe had already become dominated by conifers, with deciduous angiosperms a minor component. During the Quaternary, coniferous species and genera were lost from the European flora, leaving a few species and genera of angiosperms as the dominant trees. Cold, dry, continental climate during the glaciations caused the extinction of conifers; deciduous trees apparently survived these climatic conditions in pockets of favorable habitat in the eastern Mediterranean region. In eastern North America, in contrast, temperate deciduous forests are quite similar to the forests that were present in the late Tertiary. During the Quaternary, relatively few extinctions occurred, although deciduous angiosperms were displaced from the Appalachian mountains, surviving in small populations in the lower Mississippi valley or on the southern coastal plain. Coniferous forests dominated by spruce grew in the Great Plains, and forests dominated by pine grew on the southern part of the Atlantic coastal plain. At the opening of the Holocene, and presumably at the beginning of all the previous interglacials, tree distributions changed dramatically as temperate species rapidly extended their ranges northward. Range boundaries have continued to change throughout the Holocene, as expansions and contractions of range have occurred as the result of climatic change. Quaternary climatic history caused dramatic changes in the forests of both areas, indicating that modern species distributions can no longer be considered relicts of Tertiary distributions. Throughout the Quaternary, species ranges have changed in response to changes in regional climate; many forest communities are of recent origin, having received their present complements of tree species within the last 5,000 years. Forest communities in Eastern North America and in Western Europe as well have been invaded repeatedly during the Holocene by forest species expanding from refuges far to the south.

Temperate deciduous forest grows over a wide area of eastern North America. The forest is rich in numbers of species, especially the mixed mesophytic forest communities of the southern Appalachians. These forests were traditionally compared with forests of the Tertiary Period (Reid, 1935; Braun, 1947, 1950; Campbell, 1982), when the so-called Arcto-Tertiary geoflora was supposed to have been widespread throughout the northern hemisphere (Chaney, 1944). Reid (1935) and Chaney (1944) believed that severe climate during the Quaternary Period eliminated the Arcto-Tertiary geoflora entirely from many regions, such as western North America, while in others, such as western Europe, extinctions eliminated all but a few species and genera. The modern deciduous forests were thus seen as remnants of an originally widespread, uniform vegetation.

The geoflora concept has been challenged recently by Wolfe (1978, 1979) who argued that a uniform broad-leaved forest never existed. His analysis of the paleobotanical data shows that Tertiary floras were diverse, with evergreen gym-

nosperms such as *Sequoia* dominant in some regions, and evergreen angiosperms present elsewhere. He argued that a number of major climatic changes occurred during the Tertiary, especially during the Oligocene; these changes led to local changes in abundances of various components of the Tertiary flora. Thus changes of climate caused local adaptations rather than migrations of intact plant communities from one latitude to another as Chaney hypothesized (Wolfe, 1979). Before the end of the Tertiary Period, the forests of western United States were already dominated by conifers. A similar change had also occurred in Europe, where Pliocene floras contained many genera and species of conifers (Wolfe, 1979). *Sequoia* was the dominant tree in some regions (Traverse, 1982).

Mixed coniferous-deciduous forest in Europe during the Pliocene is a new interpretation of forest history that stands in marked contrast to the traditional view. The traditional view held that deciduous forest persisted in Europe into the early Quaternary Period, when increasing se-

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verity of climate caused extinctions of many angiosperm trees (Tralau, 1973; Campbell, 1982). In contrast, Wolfe emphasized that coniferous species and genera were the important losses from the European flora during the Quaternary. The Taxodiaceae, for example, once so important in the Black Sea region, were eliminated entirely (Traverse, 1982). The angiospermous genera that remain today represent differential survival of one component of what had been mixed coniferous-deciduous forest (Wolfe, 1979). Wolfe pointed out that Europe today has the type of climatic regime that elsewhere in the world supports mixed coniferous forest; the dominance of deciduous species in the region today is therefore anomalous. Eastern North America, in contrast, has a climatic regime typical of deciduous forest regions. Today it supports forests dominated by deciduous angiosperms, just as it did during the late Tertiary (Wolfe, 1979).

The Quaternary pollen record adds a useful perspective to these differing views of the origin of the European deciduous forest and the relationship of deciduous forests in eastern North America and Europe. First, the Quaternary record shows that dramatic changes in the geographical ranges of forest species occurred during the Quaternary. We can no longer speak of Tertiary forest "remaining" in a region throughout the Quaternary, because many tree species were repeatedly displaced during the Quaternary from the geographical region where they now occur. For this reason, *modern geographical ranges cannot be used to identify the locations of "relict" Tertiary forests*. Second, the response of temperate forest trees to Quaternary climatic change was individualistic, supporting Wolfe's contention that *Tertiary floras would not have migrated as units in response to climatic events*. Third, the extinctions that occurred during the Quaternary, especially the differential extinctions of conifers and deciduous angiosperms document that *the differential severity of Quaternary climate affected forests differently in Europe and North America*. These three factors contributed to the make-up of the modern temperate forest floras of North America and Europe.

In considering how Quaternary climate changed Pliocene floras into modern floras, the two phases of Quaternary climate, glacial climates and interglacial climates, are important. The two phases appear to have had quite different effects. Glacial phases, i.e., times when ice sheets were more extensive than at present, com-

prised about 90 percent of the time during the Quaternary period. During these long, cold intervals, temperate species survived in small populations that were susceptible to extinction. The severity of the climate, both in terms of average temperature, continentality, and drought; the extent of geographical displacement of plant species; the sizes of populations; and the community composition of forests in refuge areas; all had an effect on the probability of extinction for individual species.

Interglacial intervals comprised a much smaller proportion (about 10 percent) of the Quaternary period. They were characterized by climates similar to those of today, which in Europe and eastern North America seem to bear a general resemblance to late-Tertiary climate. Each interglacial was short, lasting only 10,000 to 15,000 years, and began and ended with a sudden, major climatic change (Emiliani, 1972; Broecker & Van Donk, 1970). The interglacials, although favorable for survival and population expansion of temperate forest trees both in Europe and eastern North America, were times of vegetational instability. During interglacials the geographical distributions of temperate species shifted many hundreds of kilometers, and the composition of forest communities changed rapidly.

GEOLOGICAL EVIDENCE OF EVENTS DURING THE QUATERNARY PERIOD

The exploration of the deep sea by geologists in the last twenty years has led to a new understanding of Quaternary events, revolutionizing our thinking about the time scale of glaciation. Many marine cores include sediment extending through the entire Quaternary Period. Previously, four major glaciations were recognized within the Quaternary. We now know that there were 18 or 20 glaciations during the last 2 million years, the time interval now assigned to the Quaternary. Each of these glacial cycles lasted about 100,000 years (Hays et al., 1969). Figure 1 shows oxygen-isotope paleoclimatic records for the last 800,000 years. The climatic events are well dated: the last interglacial (the earliest part of stage 5) started 125,000 years ago, lasted about 15,000 years, and ended with a sharp decline in temperature that initiated the last glaciation. Warm conditions returned, followed by a cold period, then a short warm interval. Seventy thousand years ago a long cold interval (stages 2-4) began, which culminated in the glacial maximum 18,000 to 20,000 years ago (Broecker & Van Donk, 1970).

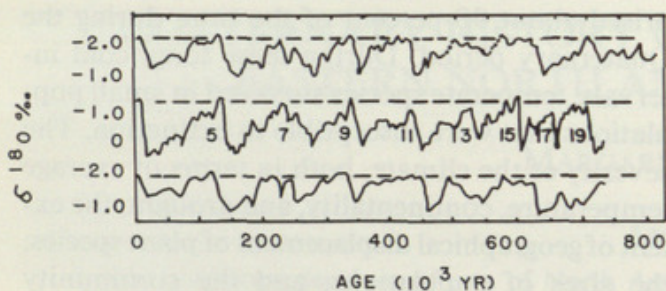


FIGURE 1. Oxygen-isotope ratios in three deep-sea cores, plotted against time. The low points on the curves represent periods when glaciers were expanded, the high points, periods when glaciers were contracted. Interglacial stages are indicated by Arabic numerals 5, 7, 9, etc., but glaciers were as small as today (isotope curves above dashed lines) for only short periods during the Quaternary. The three cores record nine glacial-interglacial cycles during the last 800,000 years (modified from Johnson, 1982).

A number of aspects of this record are important to vegetation history. The first is that changes occurred much more rapidly than previously supposed. We had imagined four rather leisurely glacial-interglacial cycles, each lasting several hundred thousand years, with long interglacials equal to, or longer than, glacial periods. The newer data indicate that interglacial periods were brief, 10,000 to 15,000 years in length, while the glacial intervals were much longer, lasting about 100,000 years. During glacial intervals there were fluctuations of climate culminating in a severe, but relatively brief glacial maximum and then a rapid, large temperature rise into the next interglacial (Broecker & Van Donk, 1970). The present interglacial, the Holocene, is typical in length for an interglacial, meaning that events recorded in Holocene sediments can be used as an analog for events during previous interglacials.

FULL-GLACIAL VEGETATION OF NORTH AMERICA

Cores of lake sediment have yielded detailed information about vegetation changes during the glacial-interglacial cycles. Sediments can be sampled at close intervals, a few hundred years or even decades apart, and pollen assemblages for the last 50,000 years can be dated by radiocarbon. Pollen percentage assemblages can be compared with assemblages in surface samples, collected from different vegetation regions. More than 1,800 surface samples are available for comparison from different parts of eastern United States and Canada (Overpeck & Webb, 1983).

These can be compared with quantitative samples of modern forest to show how the percentages of pollen relate to the percentages of growth stock volume in the modern forest (Delcourt et al., 1983). In addition, pollen concentrations in sediment can be corrected for the rate at which the sediment matrix has accumulated, and thus the actual numbers of pollen grains deposited every year on the sediment surface can be determined (Davis & Deevey, 1964). Comparisons of modern pollen deposition rates and forest records show that pollen deposition is proportional to the density of source trees within a 15 km radius of the deposition site (Davis et al., 1973; Webb et al., 1981). Although imprecisions arise because pollen grains are produced in different amounts by different species and are transported different distances, we nevertheless can use pollen to obtain not only a list of the flora, but also roughly quantitative information about the abundances of different kinds of trees. Macrofossil studies are used increasingly in combination with pollen studies to provide species identifications and definitive proof of the local presence of particular plant species (e.g., Watts, 1979; Davis et al., 1980).

Eastern United States is shown in Figure 2, with black dots indicating sites where sediments are securely dated by radiocarbon at 18,000 to 20,000 years before present, when the ice stood at its maximum. Most of these sites have been studied within the last 15 years.

The new paleobotanical data demonstrate that the vegetation during the last glacial period was very different from modern vegetation (Watts, 1980, 1984). Furthermore, most of the plant communities were dissimilar from any plant communities that are widespread today. Contrary to previous thinking, the latitudinal zonation of modern forest communities was not merely displaced southward in response to displacement of climatic zones (Martin, 1958). Instead a very different series of vegetation types developed, without modern analogs. South of the glacial boundary in New York and in Pennsylvania there was a belt of tundra, which might have been 100 to 200 km wide, extending southward at high elevations along the mountains (Watts, 1979; Maxwell & Davis, 1972). Treeline was at about 1,000 m elevation in western Maryland (Watts, 1979).

Along the Atlantic coastal plain there was coniferous forest, with *Pinus*, *Picea*, *Abies*, and *Larix* (Watts, 1984; Whitehead, 1973, 1981). *Picea*

was common in the north, but farther south, it became quite rare and *Pinus* was dominant. The pollen assemblages from South Carolina leave it ambiguous whether there was dense forest or semi-open vegetation (Whitehead, 1973; Watts, 1983), but it is clear that deciduous trees were rare or absent. On the southern coastal plain and piedmont as far south as northern Florida, *Pinus* was the dominant tree (Watts, 1970, 1983; Watts & Stuiver, 1980). There were only trace quantities of deciduous tree pollen at sites north of northern Florida (Watts, 1970, 1980; Watts & Stuiver, 1980; Whitehead, 1973, 1981).

On the western flank of the Appalachians, *Pinus* pollen dominated the fossil assemblage (Delcourt, 1979). But in the Mississippi valley, *Picea* pollen was abundant, and *Pinus* pollen was absent (Delcourt et al., 1980). Farther north and west on the Great Plains, *Picea* and *Larix* were the dominant trees. *Pinus* was present before 24,000 years ago, but it became extinct just before the last glacial maximum. Unlike the modern boreal forest, which contains both *Pinus* and *Picea*, especially in the central part of Canada, *Picea* dominated the full glacial forest in the central part of the continent, and *Pinus* was absent. There is some controversy about the presence of *Quercus*. *Quercus* pollen is present (15%) at an Illinois site at stratigraphic levels that appear to be full-glacial (Grüger, 1972). Some authors have speculated that oak grew together with spruce under a climatic regime in which seasonality was much reduced (Wright, 1982; Solomon, 1982), while others suggest that the pollen assemblage represents a tundra in which oak pollen carried by wind from a distance stands out in a pollen assemblage from unproductive tundra (King & Lindsay, 1980; Maxwell & Davis, 1972). Oak grows together with spruce today on sandy soils in southern Manitoba, producing similar pollen assemblages (West, 1961) and macrofossils indicate that oak occurred in Iowa 12,000 years ago, when spruce was still present (Baker et al., 1980).

The deciduous forest today reaches its maximum development in the Appalachian mountains. Here the community diversity is highest and many species reach maximum size and growth rate. The high diversity of this forest community has led to speculation that the southern Appalachian region was a center for the deciduous forest community, and perhaps functioned as a refuge area during the Quaternary glaciations (Braun, 1950). The fossil record dis-

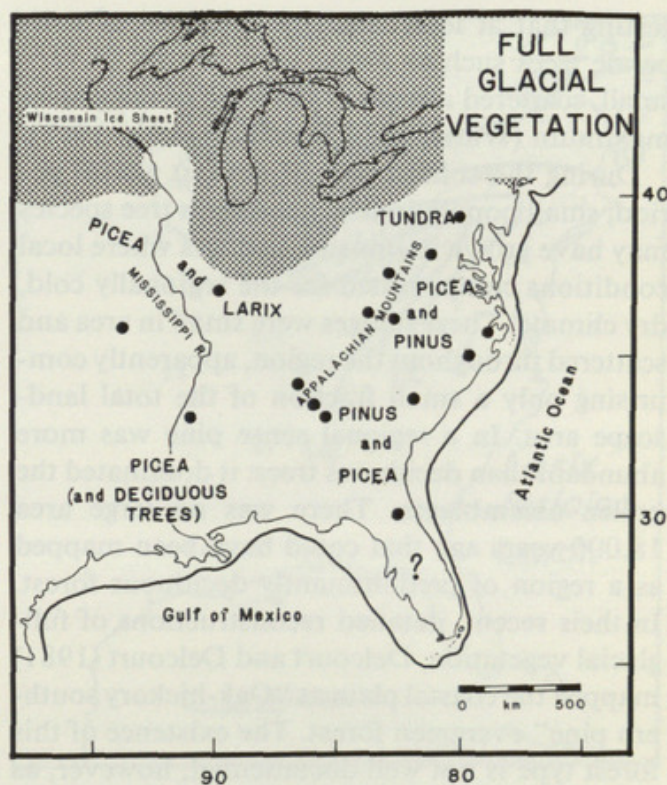


FIGURE 2. Outline map of North America, showing the extent of ice and the locations of major kinds of vegetation 18,000 years ago. Black dots indicate locations of fossil deposits dated with certainty at 18,000 to 18,500 years.

proves this idea, showing instead that deciduous forest was completely displaced from the Appalachian mountains.

Full-glacial pollen deposits are dominated by coniferous pollen; they do not show a predominance of deciduous tree pollen, such as occurs today in modern sediment from all regions dominated by deciduous forest, especially regions of mixed mesophytic forest (Delcourt et al., 1983). Deposits dated at 18,500 years before present in northern Florida contain 80 percent pollen from southern pine species. Farther south there are no records, although sand dunes and other geomorphic evidence suggest that the region was too dry for extensive deciduous forest (Watts, 1983). It is unclear where all the deciduous forest species survived the last glacial maximum. Fossil pollen, seeds and fruits from temperate deciduous trees suggest that some species grew together with spruce near Memphis in the Mississippi Valley (Delcourt et al., 1980). In northern Florida, deciduous tree pollen (mainly *Quercus* and *Carya*) makes up 15 to 20 percent of the pollen assemblage (Watts & Stuiver, 1980). Deciduous tree pollen increases in abundance rapidly at many southern sites during the late-glacial period, sug-

gesting that at least small populations of temperate trees such as *Fagus* grew nearby in very small, scattered refuges at the time of the glacial maximum (Watts, 1984; Solomon et al., 1981).

During the coldest part of the last glacial period, small populations of deciduous tree species may have grown in unusual habitats where local conditions compensated for the regionally cold, dry climate. These refuges were small in area and scattered throughout the region, apparently comprising only a small fraction of the total landscape area. In a regional sense pine was more abundant than deciduous trees: it dominated the pollen assemblages. There was no large area 18,000 years ago that could have been mapped as a region of predominantly deciduous forest. In their recent, detailed reconstructions of full-glacial vegetation, Delcourt and Delcourt (1981) mapped the coastal plain as "Oak-hickory southern pine" evergreen forest. The existence of this forest type is not well documented, however, as evidence has not been published from sites on the coastal plain except northern Florida, where it is certain that sedimentation occurred continuously during the full-glacial (Watts, 1984). Although Delcourt and Delcourt implied a higher frequency for angiospermous trees than I have suggested, they were in agreement that the full-glacial vegetation of this region was different from the modern mixed mesophytic forest of the Appalachians.

HOLOCENE FORESTS

Additional information about the history of deciduous tree species comes from records of vegetation after the ice began to retreat. In this case we have many sites, not only those in Figure 2, but also many north of the glacial boundary, where lakes and bogs are plentiful. These localities are indicated by the small numbers in subsequent figures; references are given in Davis (1981).

Pollen records were used to determine the northward movement of trees recolonizing deglaciated territory during the early Holocene (Davis, 1976, 1981). Of course pollen can be dispersed far from living populations of plants; some quantitative criterion must be developed to use pollen grains as evidence for the local arrival of trees. Modern pollen deposition rates in lakes show a relationship to geographical range limits. Accumulation rates for all species studied are higher within the range limit than outside of

it, and higher still where the species is abundant. These relationships were different for the different species that were studied, depending apparently on the production rates and dispersal tendencies of the pollen. Generally, pollen deposition rates are at least 5-fold, and in some cases 10-fold higher within the range than outside (Davis et al., 1973). On this basis I have deduced that the earliest steep rise in the rate of pollen deposition in fossil deposits, that is a ten-fold increase to levels similar to those observed in modern sediments within the range of the source species, can be used as evidence of local arrival. In a few instances where data were available only as relative counts, I used a steep rise in pollen percentages in radiocarbon-dated pollen percentage diagrams. In fitting isochrones, however, more weight was given to data points based on deposition rates.

Recognizing that pollen percentages can be as low as one percent near the geographical limit for a species, I have avoided using a particular pollen percentage value to indicate presence. Low percentage values are difficult to determine accurately because of statistical errors. Macrofossils have corroborated the interpretations in many cases (Davis et al., 1980). There are also cases where macrofossils have been found at nearby sites where population expansion occurred earliest were closest to refuge areas, and sites that show later arrivals are farther north. Maps showing northward range extensions of several major forest trees have been prepared from published pollen evidence using the criteria described above (Figs. 3, 4, 5).

Arrival times (rounded to 1,000-year intervals) were indicated on a map, and isochrones were drawn connecting points of similar age. The sites where population expansion occurred earliest were closest to refuge areas, and sites that show later arrivals are farther north. Maps showing northward range extensions of several major forest trees have been prepared from published pollen evidence using the criteria described above (Figs. 3, 4, 5).

Figure 3, for example, shows data for *Picea*. The stippled area is the region where *Picea* grows today. The small numbers on the map show "arrival times" in thousands of years before present, at each site. Isochrones have been drawn connecting points of similar age, tracing the leading edge of the expanding population. The scatter of points indicates the uncertainty of the determinations of arrival dates. Pollen percentages com-

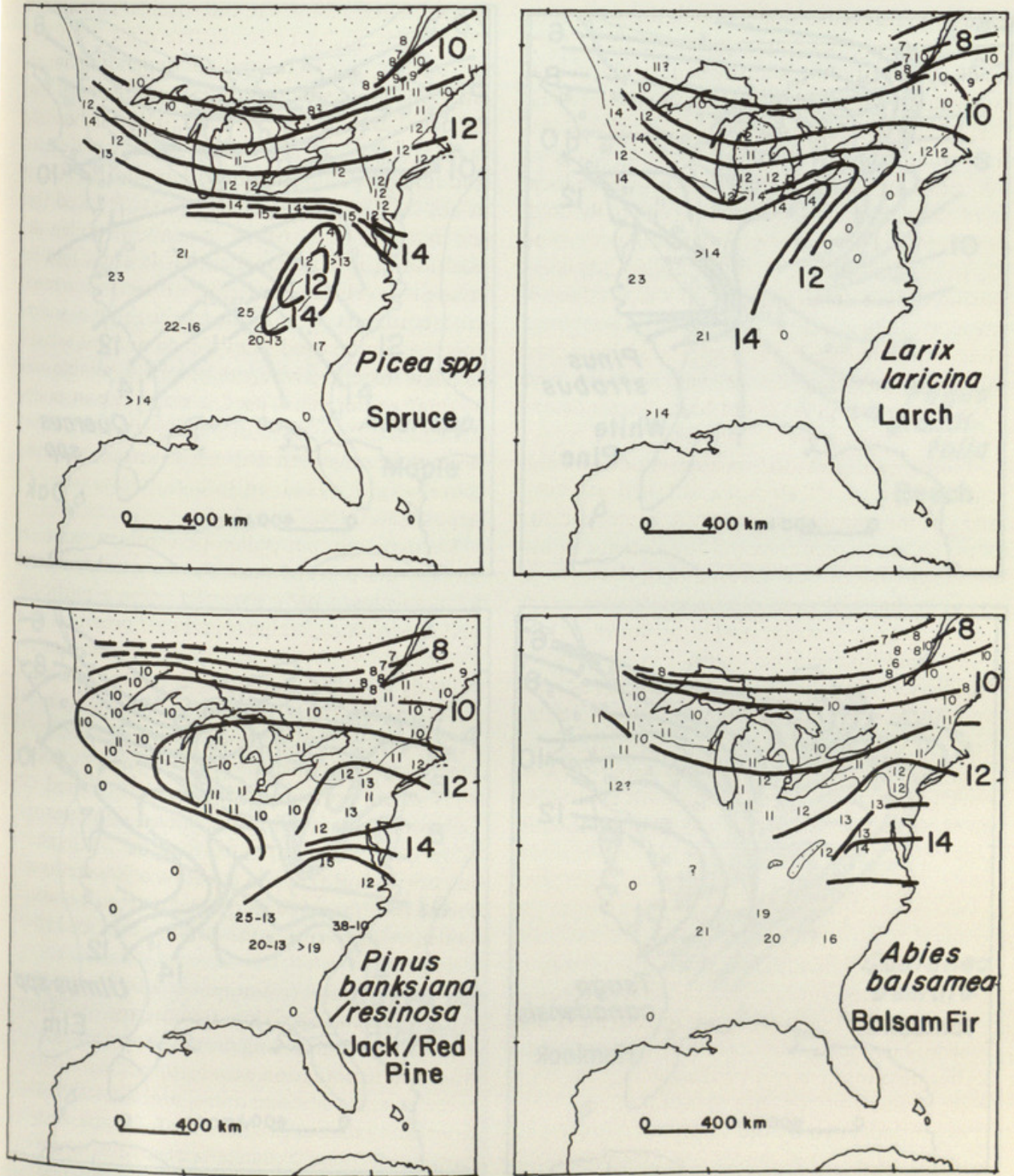


FIGURE 3. Expansion of boreal tree species northward following the retreat of the ice. Small numbers indicate arrival times in thousands of years before present at individual sites, as indicated by fossil pollen (see text). Lines connect points of similar age, indicating the location of the expanding frontier for the species at 1,000-year intervals. Shaded area is present-day range for the genus or species (from Davis, 1983).

piled from sites over a wide area have been plotted on a similar series of maps by Bernabo and Webb (1977), Webb (1981), and Huntley and Birks (1983). Pollen percentages are affected by the background of pollen produced by other

species, as well as by the amount of pollen produced by the population of interest. Nevertheless a map of percentages shows the region where the species was most abundant (given a more or less constant pollen background) and it shows the

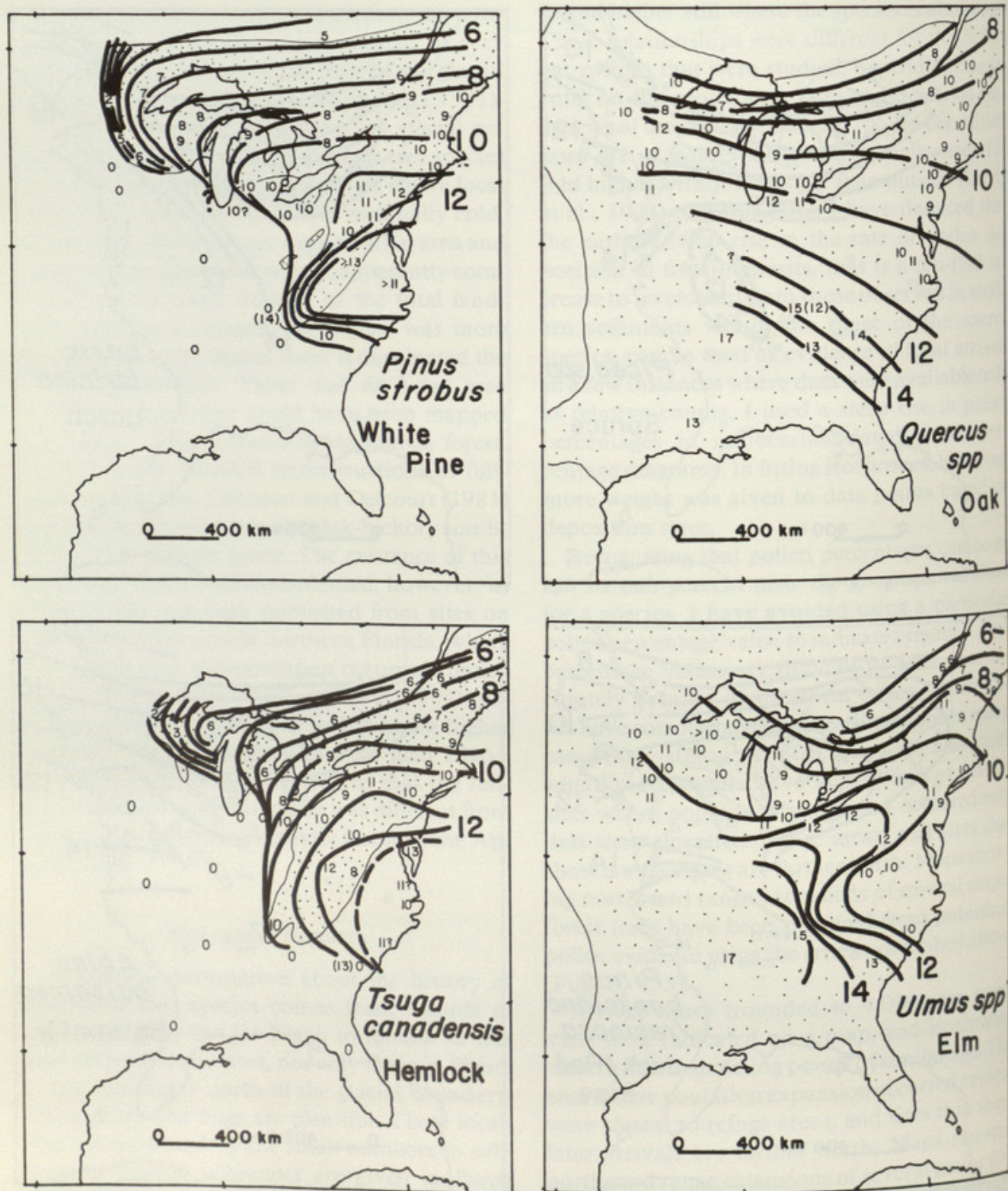


FIGURE 4. Range extension of two coniferous tree species that commonly grow in deciduous forests, and *Quercus* and *Ulmus*. Both coniferous species expanded from the eastern coastal plain, suggesting a refuge area somewhere in the vicinity. Oak and elm expanded in a northeasterly direction from refuges that were apparently located in the southern Mississippi valley region (from Davis, 1983).

retreating edge of the migrating population (Bernabo & Webb, 1977). Data showing later arrivals at northern sites than at southern sites can be used to calculate rates of range extension in meters per year. Most tree species were able to extend their ranges northward 200 to 300 meters

per year, a rapid rate for trees with long generation times. These estimates are supported by data from Europe, which suggest similar rates of expansion there (Firbas, 1949; Huntley & Birks, 1983).

In northeastern United States, far from the

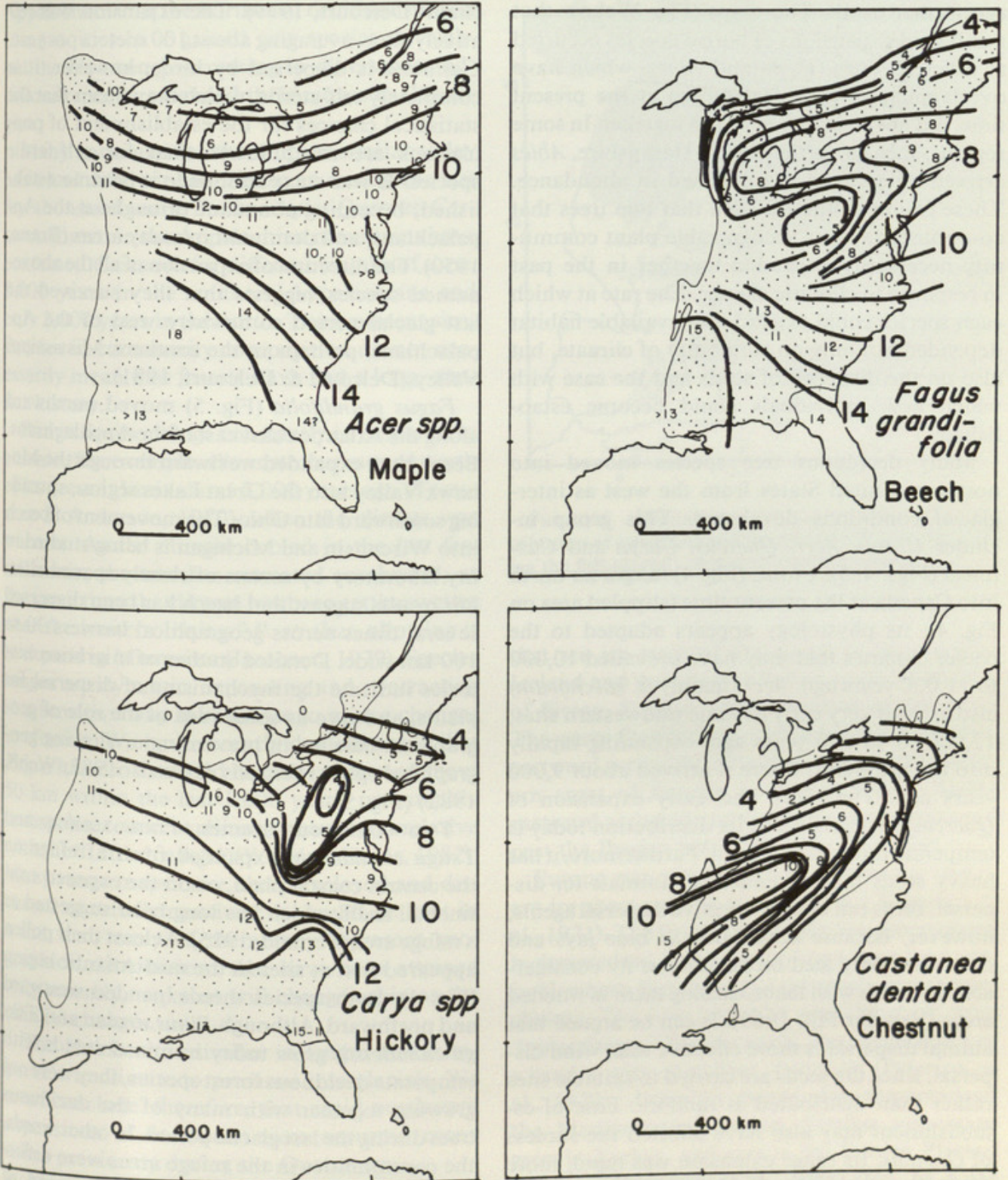


FIGURE 5. Range extension of four deciduous tree species or genera following the retreat of the ice. Chestnut was the slowest species to extend its range northward. Beech moved northward east of the Appalachians, expanding westward across the lower Great Lakes region (from Davis, 1983).

refuge areas where trees grew during the glacial maximum, the forest vegetation gained species gradually through time, increasing in diversity throughout the Holocene. Certain forest species arrived very late—*Castanea* arrived in Connecticut as recently as 2,000 years ago. Sites in the southern Appalachians, such as Anderson

Pond in Tennessee (Delcourt, 1979), were closer to refuges and much less influenced by migration histories (Solomon et al., 1981).

The replacement of *Picea* and other boreal species by deciduous forest or by mixed deciduous forest was time-transgressive, occurring 16,000 years ago in the south and 10,000 years

ago farther north. The maps (Fig. 3) show that northward expansions of boreal species occurred at different rates. *Picea* and *Abies*, which have overlapping geographical ranges at the present time, did not expand northward together. In some regions, such as northern New Hampshire, *Abies* arrived after *Picea* had declined in abundance. These results refute the idea that two trees that co-occur today in a recognizable plant community necessarily expanded together in the past in response to climatic change. The rate at which each species could expand into available habitat depended not only on suitability of climate, but also on the dispersal of seeds and the ease with which new individuals could become established.

Many deciduous tree species moved into northern United States from the west as interglacial conditions developed. This group includes *Ulmus*, *Acer*, *Quercus*, *Carya* and *Castanea* (Figs. 4, 5). *Ulmus* (Fig. 4) ranges far north into Canada at the present time (stippled area on Fig. 4); its physiology appears adapted to the cooler climates that may have prevailed 10,000 to 11,000 years ago. *Acer* (mainly *A. saccharum*) also arrived very early at some midwestern sites, 12,000 to 11,000 years ago, expanding rapidly into the northeast, where it arrived about 9,000 years ago. The rapid and early expansion of *Quercus* is surprising, as its distribution today is temperate rather than boreal. Furthermore, it has heavy seeds that depended on animals for dispersal. Birds can be very effective dispersal agents, however, because birds such as blue jays and crows not only feed on acorns but fly considerable distances with them, caching them in wooded areas (Van der Pijl, 1969). It can be argued that animal dispersal is more effective than wind dispersal, since the seeds are carried to suitable sites rather than distributed at random. Ease of establishment may also have affected the success of *Quercus*. Its range extension was rapid, more rapid than any of the deciduous trees that today grow in mature forests. Also moving from the west, but expanding more slowly, was *Carya* spp., which arrived early (10,000 B.P.) in the Midwest, even at sites close to its present northern boundary. But it was slow to cross the Appalachian mountains and arrived late in New England, only 5,000 years before present.

Castanea pollen is found in quantity near Memphis, Tennessee, as early as 15,000 years ago (Delcourt et al., 1980). It then expanded northeastward up the Appalachian mountain

chain (Delcourt, 1979). The expansion was relatively slow, averaging about 100 meters per year. *Castanea* is dispersed by birds; however, it is completely self-sterile, which may mean that the statistical chances for the establishment of populations are much lower than for self-fertile species. It was successful once it became established, becoming abundant throughout the Appalachians, and dominant in many areas (Braun, 1950). The direction of expansion of all the above-named species suggests that they survived the last glacial period somewhere west of the Appalachians, perhaps in the southern Mississippi Valley (Delcourt & Delcourt, 1981).

Fagus grandifolia (Fig. 5) moved northward along the Atlantic coast east of the Appalachians. Beech then expanded westward through the Mohawk Valley into the Great Lakes region, spreading southward into Ohio. The movement of beech into Wisconsin and Michigan is being studied in my laboratory by means of closely spaced sites; our results suggest that beech has been dispersed several times across geographical barriers 30 to 100 km wide. Detailed studies of migration histories indicate the mechanisms of dispersal for plants, and give us some idea of the role of geographical barriers in preventing or slowing geographical spread (Woods & Davis, 1982; Webb, 1982) (Fig. 5).

Two coniferous species, *Pinus strobus* and *Tsuga canadensis*, expanded from a refuge on the central coastal plain, or on the exposed continental shelf, which has long been suggested as a refuge area (Fernald, 1925). At least their pollen appeared first at sites in the mid-Atlantic region (Fig. 4). Both species then expanded westward and northward. Although *Pinus strobus* and *Tsuga canadensis* grow today in mixed stands with temperate deciduous forest species, they were not growing together with many of the deciduous trees during the last glacial period. In other words, the communities in the refuge areas were different from modern communities, composed of small subsets of the species characteristic of the diverse "mesophytic forest." This is an important point, because trees were exposed to competition from species co-occurring in refuge areas for about 50,000 years. The makeup of the communities is important when the evolutionary history of species and coadaptations between species are considered. In contrast, deciduous tree species have been growing in the Appalachians where we study them today for fewer than 12,000 years, and for less than half that time in the northern

Appalachians. In the northeast, deciduous forest communities have gained new species throughout the Holocene, increasing in diversity as additional species extended their ranges to this region.

Many communities that appear similar today have had different histories even during the last few thousand years. For example, *Quercus-Castanea* forests in the Appalachian mountains have contained *Castanea* for different lengths of time, 8,000 years in the southern Appalachians and only 2,000 years in the north. Thus, present similarities in community composition do not necessarily mean that forests have had a long, similar history.

Range limits change constantly in response to climate. For example, *Pinus strobus* moved north of its present limit about 5,000 years ago (Terasmae & Anderson, 1970), when the climate was warmer. As the climate cooled in the last few millennia, the tree has retreated southward; at the same time the climate has been getting wetter near the western limit of *P. strobus*, allowing it to expand in Minnesota (Jacobson, 1979). Species ranges are dynamic, expanding and contracting in a sensitive way in response to minor changes in climate. We have recently demonstrated that *Fagus* has also extended its range westward about 50 km within the last 1,000 years, presumably in response to recent climatic changes correlative with the "Little Ice Age" (Woods & Davis, 1982). Many range maps have been displayed and discussed at this Symposium; it should be kept in mind that many of them may represent geographical distributions that have been attained only recently, and that will change again as the climate changes.

The Holocene is similar in length and comparable in climate to previous interglacials. We must conclude, therefore, that rapid northward expansions of deciduous trees occurred many times previously during the Quaternary.

DECIDUOUS FORESTS OF EUROPE

A detailed knowledge of Pleistocene biotic history is available from Europe, where interglacial deposits have been studied in detail, making it possible to observe the timing of changes in the floristic composition of temperate forest communities. Recent work has indicated that extremely severe climatic conditions prevailed in Europe during the glaciations. Reconstructed ocean surface temperatures 18,000 years ago show

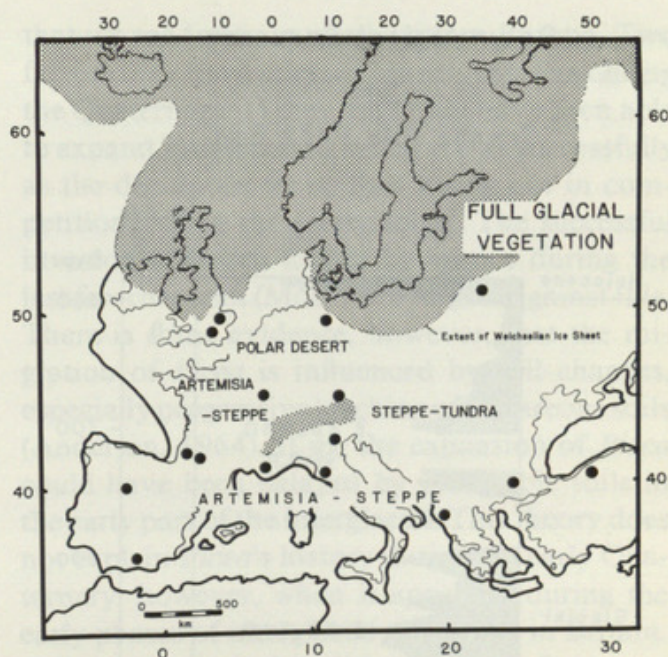


FIGURE 6. Outline map of Europe, showing the distribution of ice sheets and major vegetation types during the full-glacial, 18,000 to 20,000 years ago.

that sea ice covered the ocean as far south as Iceland and polar water flowed south to the Bay of Biscay. Surface water temperature fell 10°C. The warm Gulf Stream, instead of flowing north-eastward across the Atlantic and along the western coast of Europe, as it does today, flowed eastward across the mid-Atlantic and southward past the Iberian peninsula (Climap, 1976).

Europe was without forest between the northern ice sheet and the Alps (Van der Hammen et al., 1971). The floor of the North Sea was exposed by lowered sea levels, providing a large land area in Northern Europe and a relatively continental climate. The low countries and Denmark were "polar desert." Farther south, the vegetation was a tundra with abundant *Artemisia* (Peterson et al., 1979). *Artemisia* steppe also characterized the Mediterranean region. Similar vegetation grew in Turkey, and east to the Iranian Plateau (Van der Hammen et al., 1971) (Fig. 6). Cold and dry climate is indicated.

Pollen diagrams from interglacial deposits in Europe are similar to Holocene pollen diagrams; they show the successive arrivals of different species as trees extended their ranges northward from glacial refuges. At sites in central and northern Europe, interglacial pollen diagrams begin with a boreal flora of *Pinus* and *Betula*, and end with *Pinus*, *Betula*, and *Picea*; in the intervening temperate zone the diversity of the flora gradually increases as more and more deciduous tree

Palynological Sequence of Grande Pile Peat Bog

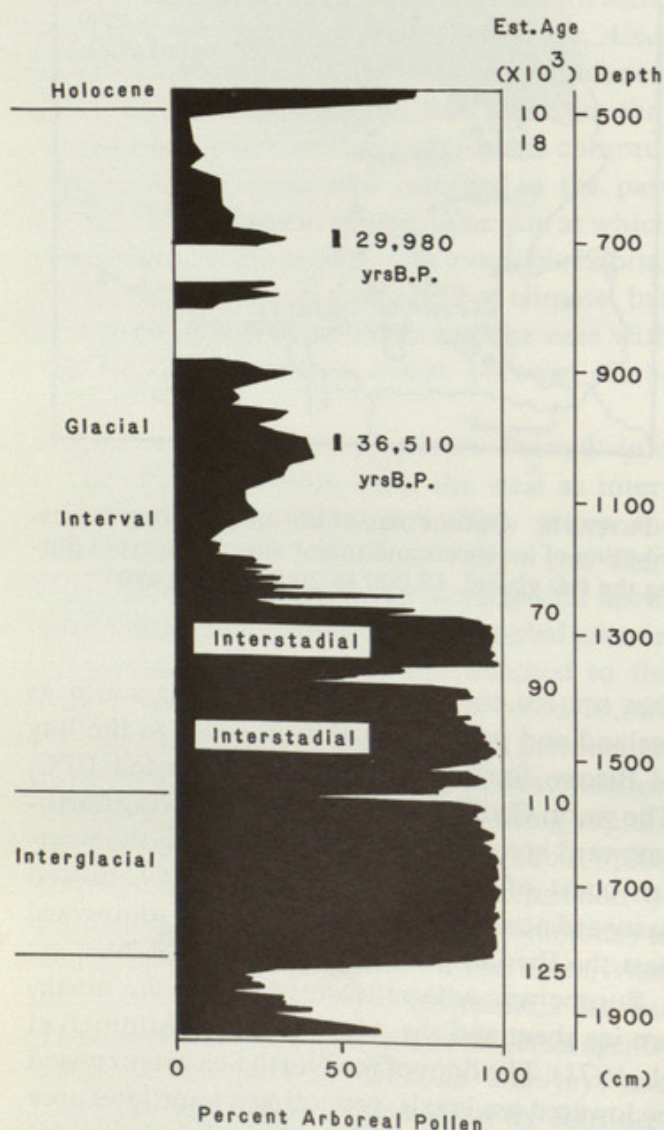


FIGURE 7. Percentages of tree pollen in the Grand Pile pollen diagram (from Woillard, 1978).

species appear. The younger interglacials and the Holocene are marked by the very late arrival of both *Carpinus* and *Picea*; these genera appeared earlier in the older interglacials.

The Grand Pile Peat Bog in northern France is a recently investigated site that is remarkable because it provides a continuous record of vegetation during interglacial and glacial intervals. A continuous sedimentary sequence includes the Holocene, the time of the last glaciation, the last interglacial and the previous glacial maximum, which occurred about 130,000 years ago (Woillard, 1978). Figure 7 is a much simplified pollen diagram that shows the percentage of tree pollen within the pollen total. The penultimate glacial maximum 130,000 years ago is marked by low

percentages of tree pollen. Tree pollen was abundant during the last interglacial, which extended from 125,000 to 110,000 years ago. The interglacial terminated with a change in climate 110,000 years ago that is also recorded in deep sea sediments. Forest disappeared from the vicinity of Grand Pile and was replaced by an *Artemisia* steppe. One hundred thousand years ago, the climate became warmer again, and the forest returned. A forest succession similar to the Holocene is indicated by the pollen diagram, as all the forest species, including *Picea*, returned. *Picea* had had a late arrival during the preceding interglacial interval, but it must have been growing nearby during the subsequent cold stadial, because it was available for rapid recolonization when the climate warmed 100,000 years ago. A second cold period began about 90,000 years ago. Again, the forest was replaced by *Artemisia* steppe, and again, about 10,000 years later, the forest returned, including all tree species. Then the climate became cold again 70,000 years ago, and a long cold period ensued, culminating in the glacial maximum 18,000 years ago. During this long period, *Artemisia* steppe prevailed locally and the deciduous forest and *Picea* apparently retracted to distant refuges.

In her classic paper, Reid (1935) emphasized the importance of east-west trending mountains in the extinction of deciduous forest species in Europe. She believed that the glaciers were accompanied by cold climate, which caused extinctions of temperate trees everywhere north of this mountain barrier. Only those species that reached refuges in the Mediterranean survived. Reid's discussion emphasized the onset of glaciation as a time of extinction. The Grand Pile data imply, however, that although forest disappeared locally in response to cold climate at the onset of glaciation, all species were available to return as soon as the climate became warm again. A more important extinction time was the very long and very severe cold period that included the glacial maximum (70,000 to 16,000 years ago), rather than the first onset of cold 100,000 years ago that accompanied the development of continental ice sheets. It was during long cold periods, lasting 50,000 years or more, that trees like *Picea* became locally extinct, surviving only in refuge areas far to the east. When the next interglacial, the Holocene, began, *Picea* was slow to return to western Europe—in fact it never did return during the Holocene to the vicinity of Grand Pile.

Distributions of deciduous trees expanded in

Europe during the Holocene just as they did in North America. It seems likely that somewhere on the southern slopes of the Alps and/or in the Balkans there were habitats during the full-glacial that were suitable for deciduous trees (Van der Hammen et al., 1971); probably small populations were scattered about in pockets of favorable environment. Maps of Holocene pollen frequencies in Europe show that many deciduous tree species expanded first in southeastern Europe and then moved westward (Huntley & Birks, 1983). The gradient in ocean surface temperature across the Mediterranean (Climap, 1976) may have been correlated with more favorable temperatures on land as well.

The succession of interglacial deposits in Europe record the progressive extinctions of tree species during the Quaternary (West, 1970; Wolfe, 1979). Different interglacials witnessed the development of different communities of deciduous trees, depending on which species arrived in the British Isles. Floristic differences among the British interglacials were also caused by evolutionary changes. West (1980) suggested that *Picea*, for example, which appeared in Britain during early interglacials, must have been migrating to the British Isles from refuge areas in western Europe. In the later interglacials there was a gradual loss of biotypes that were able to survive in refuges in western Europe. *Picea* appeared later and later in successive interglacials, and during the Holocene it failed entirely to expand into Great Britain.

Additional evidence for evolutionary changes is provided by *Corylus* pollen percentages in the successive British interglacials (West, 1980). In the early interglacials of East Anglia, *Corylus* pollen percentages are low. Later in the Quaternary it became more abundant, and finally, in the Holocene, it arrived early and built up a large population before other deciduous trees arrived. West suggested that *Corylus* may have adapted to interglacial conditions. Whereas *Picea* was less and less successful, and *Tsuga* became extinct altogether, *Corylus*, a deciduous angiosperm, became more and more abundant.

DISCUSSION

Extinction rates in Europe during the Quaternary were higher for coniferous trees than for temperate deciduous angiosperms (Wolfe, 1979). The preferential elimination of conifers documented in Quaternary deposits led to the development of the predominantly deciduous forest

that we see today in northwestern Europe. Two factors may have selected against conifers during the Quaternary: 1) they might not have been able to expand into forested landscape as successfully as the deciduous trees, thus losing out in competition during the interglacials. The successful invasion of beech forests by spruce during the last few centuries (Moe, 1970) argues against this. There is some evidence, however, that the migration of *Picea* is influenced by soil changes, especially progressive leaching of calcareous soils (Andersen, 1964). If so, the expansion of *Picea* could have been delayed by unsuitable soils in the early part of the interglacial. This theory does not explain *Picea*'s history during the early Quaternary, however, when it appeared during the early phases of interglacial sequences in Britain.

2) Alternatively, conditions in the refuge areas might have selected against conifers. The extinction of western European populations of *Picea* mid-way through the Quaternary (West, 1980) may be related to the very severe conditions that existed during the glacials in the western Mediterranean region. A very dry and probably continental glacial climate may have selected against spruce, even though interglacial climates, like the Holocene, were favorable for conifers. It would be a mistake to study the present climate and try to use it to explain all aspects of the distribution of the modern flora (Wolfe, 1979). Instead, the longer intervals of time during which the climate was very different must be taken into account, i.e., the glacial intervals, which comprise nine-tenths of the Quaternary Period. During these longer intervals, extinctions of conifers occurred, although many temperate angiosperms were able to survive.

The Quaternary fossil record has further implications for the geoflora concept. Geofloras are forest communities that were believed to persist, unchanged in composition, over many millions of years. In contrast, Holocene forest communities are ephemeral, composed of newly immigrated species, and easily invaded by additional species. The deciduous trees growing in our forests that have persisted from Tertiary time are aggressive species that have been able to move rapidly onto deglaciated territory. Any species that has not been able to disperse its seeds easily, to extend its range, and to penetrate forest communities, has become very rare or extinct. Wolfe (1979) compares our tree flora to secondary species in the forests of eastern Asia. The nature of modern communities calls into question whether the concept of closely co-adapted com-

munities such as "geofloras" is appropriate to describe Tertiary forests.

The deciduous forest communities of eastern North America and western Europe remain a grab-bag of species with characteristics that enabled them to survive during the Quaternary in the very unstable environment of interglacials, when rapid dispersal and colonization abilities were selected for. During the glacials, dry and continental climate exerted a different kind of selective pressure. Campbell (1982) has compared the tree floras of Europe and eastern North America, invoking these factors to explain some of the differences. The modern deciduous forests are a product of Quaternary history over the last two million years. Although on the generic level the forests of eastern North America bear a similarity to Tertiary forests of the same region, the marked changes that have occurred in Europe caution us against using the present forest as a model for the forests of the Tertiary.

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